

CS Topics: Mobile Application Development

Interfaces & Anonymous Inner Classes Practice

Essential Questions:

- *What is an anonymous inner class?*
- *In which situations would a programmer use a defined inner class over an anonymous inner class and vice versa?*

For these tasks, it is recommended to use Java documentation for different interfaces and classes implemented to complete these assignments. If you are having trouble finding or understanding the documentation on the website, try using Google to narrow down your search or find alternative explanations on the Oracle website. **Only when you have exhausted all other resources may you use AI to assist you in creating an explanation.**

1. Write a program that allows users to enter input on their keyboard and print out whether they type vowels or consonants. You may assume that the user will only type in lowercase characters. The program should use an anonymous inner class.
2. Write a program that will print out the mouse's current location within the window (you can get this by calling a method on the [MouseEvent](#)). The program should employ an anonymous inner class. Make sure that you add the listener to the **JPanel**. (Hint: Use **addMouseMotionListener** and the [MouseMotionListener](#) interface. Refer to our **KeyListener** demo, which will assist you. I suggest adding a **frame.setSize(300, 300)** after you instantiate the **JFrame** to resize the window to make it more easily accessible.)
3. Create a **Calculator** interface with two methods: **add(int a, int b)** and **subtract(int a, int b)**. Remember, the interface should have abstract methods (only method headers). Implementation for **add** and **subtract** will be written in the anonymous class that employs the interface. After writing the interface, write a program that uses an anonymous inner class to implement this interface and perform addition and subtraction operations on its parameter values. You should be able to test the program by printing the **result1** and **result2** from the following statements:

```
int result1 = calculator.add(5, 3);  
int result2 = calculator.subtract(8, 4);
```

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4. Add a code segment to the following program to create a GUI application with a button and a text field. When the user clicks the button, the text field should display "Button Clicked." Use an anonymous inner class to handle the button click event.

Java

```
import javax.swing.*;
import java.awt.event.*;

public class Task4 {

    public static void main(String[] args) {

        JFrame frame = new JFrame("Task 4");
        JTextField textField = new JTextField();
        textField.setText("Button NOT Clicked");

        // 1. Declare and instantiate a JButton, named "button"

        // 2. Apply the addActionListener method to the JButton

        /* 3. Use an anonymous inner class to add and define an
         * ActionListener to the button's addActionListener call
         */

        // Your code segment goes here **/

        frame.setLayout(new java.awt.FlowLayout());
        frame.add(button);
        frame.add(textField);
        frame.setSize(300, 100);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```